

Project Name: New project

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BILL OF MATERIAL

Controller

	Reference	TM221CE24R
	Description	TM221CE24R (screw) 14 digital inputs, 10 relay outputs (2 A), 2 analog inputs, 1 serial line port, 1 Ethernet port, 100-240 Vac power supply controller with removable terminal blocks.
	Power supplied to the IO bus	5V: 520 mA / 24V: 160 mA

Cartridge

	Reference	TMC2AI2
	Description	TMC2AI2 TMC2 cartridge with 2 analog voltage or current inputs (0 - 10V, 0-20mA, 4 - 20mA), 12 bits.

HARDWARE CONFIGURATION

MyController - TM221CE24R

Digital Inputs

Used	Address	Filtering	Latch	Run/Stop	Events	Priority	Subroutine
X	%I0.0	3 ms			Not Used		
X	%I0.1	3 ms			Not Used		
X	%I0.2	3 ms			Not Used		
X	%I0.3	3 ms			Not Used		
X	%I0.4	3 ms			Not Used		
X	%I0.5	3 ms			Not Used		
X	%I0.6	3 ms			Not Used		
X	%I0.7	3 ms			Not Used		
	%I0.8	3 ms			Not Used		
	%I0.9	3 ms			Not Used		
	%I0.10	3 ms			Not Used		
	%I0.11	3 ms			Not Used		
	%I0.12	3 ms			Not Used		
	%I0.13	3 ms			Not Used		

Digital Outputs

Used	Address	Status Alarm	Fallback value	Used by
X	%Q0.0		0	User logic
X	%Q0.1		0	User logic
X	%Q0.2		0	User logic
X	%Q0.3		0	User logic
X	%Q0.4		0	User logic
X	%Q0.5		0	User logic
X	%Q0.6		0	User logic
X	%Q0.7		0	User logic
X	%Q0.8		0	User logic
X	%Q0.9		0	User logic

Analog Inputs

Used	Address	Type	Scope	Range	Filter	Sampling
	%IW0.0	0 - 10 V	Normal	0-1000	0	
	%IW0.1	0 - 10 V	Normal	0-1000	0	

Fast Counters

Used	Address	Input	Configured	Preset	Double Word
	%FC0	%I0.2	NotUsed	0	
	%FC1	%I0.3	NotUsed	0	
	%FC2	%I0.4	NotUsed	0	
	%FC3	%I0.5	NotUsed	0	

High Speed Counters

Used	Address	Type
	%HSC0	Not Configured
	%HSC1	Not Configured
	%HSC2	Not Configured
	%HSC3	Not Configured

ETH1

Device name:	M221
IP Mode:	Fixed
IP address:	0.0.0.0
Subnet mask:	0.0.0.0
Gateway address:	0.0.0.0
Transfer Rate:	Auto
Security Parameters:	Programming protocol disabled
	Auto discovery protocol disabled
	Modbus server disabled
	EtherNet/IP protocol disabled

SL1 (Serial line)

Physical Settings

Device:	None
Baud rate:	19200
Parity:	Even
Data bits:	8
Stop bits:	1
Physical medium:	RS-485
Polarization:	No

Protocol Settings

Protocol:	Modbus
Response timeout (× 100 ms):	10
Time between frames (ms):	10
Transmission mode:	RTU
Addressing:	Slave
Address:	1

Cartridges

TMC2AI2

Analog Inputs

Used	Address	Type	Scope	Range	Filter	Sampling
X	%IW0.100	4 - 20 mA	Normal	4000- 20000	0	
	%IW0.101	4 - 20 mA	Normal	4000- 20000	0	

SOFTWARE CONFIGURATION

Constant Words

KW

Allocation: Automatic

Allocated: 0

Used	Equ Used	Address	Symbol	Value	Comment
------	----------	---------	--------	-------	---------

KD

Allocation: Automatic

Allocated: 0

Used	Equ Used	Address	Symbol	Value	Comment
------	----------	---------	--------	-------	---------

KF

Allocation: Automatic

Allocated: 0

Used	Equ Used	Address	Symbol	Value	Comment
------	----------	---------	--------	-------	---------

Network Objects

Input Assembly (Ethernet/Ip)

Used	Address	Symbol	Fallback value	Comment
------	---------	--------	----------------	---------

Output Assembly (Ethernet/Ip)

Used	Address	Symbol	Comment
------	---------	--------	---------

Input Registers (Modbus Tcp)

Used	Address	Symbol	Fallback value	Comment
------	---------	--------	----------------	---------

Output Registers (Modbus Tcp)

Used	Address	Symbol	Comment
------	---------	--------	---------

Digital inputs (IOScanner)

Used	Address	Channel	Symbol	Comment
------	---------	---------	--------	---------

Digital outputs (IOScanner)

Used	Address	Channel	Fallback value	Symbol	Comment
------	---------	---------	----------------	--------	---------

Input registers (IOScanner)

Used	Address	Channel	Symbol	Comment
------	---------	---------	--------	---------

Output registers (IOScanner)

Used	Address	Channel	Fallback value	Symbol	Comment
------	---------	---------	----------------	--------	---------

Software Objects

Timers

Allocation: Automatic

Allocated: 2

Used	Address	Symbol	Type	Retentive	Dynamic Preset	Time Base	Preset	Comment
X	%TM0		TON			1 s	180	
X	%TM1		TON			1 s	300	DELAY_STAIRT

Counters

Allocation: Automatic

Allocated: 0

LIFO/FIFO Registers

Allocation: Automatic

Allocated: 0

Drums

Allocation: Automatic

Allocated: 0

Shift Bit Registers

Allocation: Automatic

Allocated: 0

Step Counters

Allocation: Automatic

Allocated: 0

Schedule Blocks

Allocation: Automatic

Allocated: 0

RTC

PID

Used	PID	Symbol	Type	Comment
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Grafcet Steps

Allocation: Automatic

Allocated: 0

PROGRAM

Behavior

Functional level:	Level 14.0
Starting mode:	Start In Run
Watchdog:	250 ms
Fallback behavior:	Fallback value
String end character:	CR (Carriage Return)

Memory consumption

A successful compilation is required to obtain memory information.

Application Architecture

Master Task

Scan mode:	Normal
POU list:	1 - POU_Inputs
	2 - POU_Scaling
	3 - Main_Program
	4 - POU_Modes
	5 - POU_Valve_Logic
	6 - POU_Outputs

Periodic Task

Period:	255 ms
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POU

Master Task

1 - POU_Inputs

Master Task

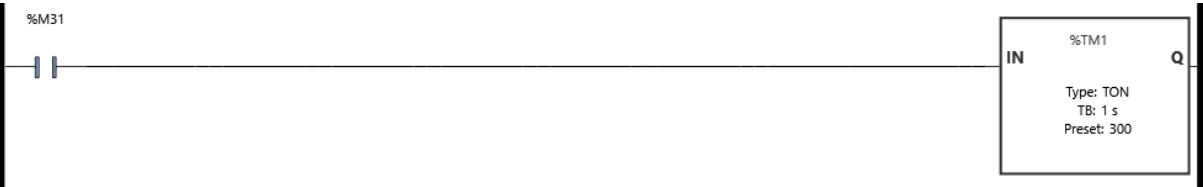
Rung0



Variables used:

%M31	POWER_DELAY_ACTIVE	
%S1	SB_WARMSTART	Indicates there was a warm start with data backup

Rung1



Variables used:

%M31	POWER_DELAY_ACTIVE	
%TM1	DELAY_START	

Rung2



Variables used:

%I0.3	DAIKIN_ALARM	
%M4	ALARM_INT	

Rung3



Variables used:

%I0.2	CHILLER_PUMP_FB	pump feedback
%M21	CHILLER_PUMP_FB_INT	

Rung4



Variables used:

%I0.4	DAIKIN_RUN_FB	Fault Signal (80-81)
%M30	DAIKIN_RUN_FB_INT	

Rung5



Legend:

1 %MW2 := %IW0.100

Variables used:

%IW0.100
%MW2 TEMP_SENSOR_RAW

Rung6



Variables used:

%I0.5 DAIKIN_MODE Operation Feedback(82-83)
%M5 DAIKIN_MODE_FB

Rung7



Variables used:

%I0.6 CIRC_PUMP_FB Water Flow Signal(70-71)
%M27 PUMP_FB_INT

Rung8



Variables used:

%I0.7 DI_VALVE_FB
%M17 VALVE_AT_CHILLER

Rung9



Variables used:

%I0.7 DI_VALVE_FB
%M18 VALVE_AT_BOILER

2 - POU_Scaling

Master Task

Rung0



Legend:

1 MF0 := ((INT_TO_REAL(%MW2) - 4000.0) / 16000.0) * 80.0 - 20.0

Variables used:

%MW2 TEMP_SENSOR_RAW

3 - Main_Program

Master Task

Rung0



Legend:

1 %TM1.Q

Variables used:

%I0.0	MAIN_START	System Start Button
%I0.1	MAIN_STOP	Emergency Stop Button
%M0	SYSTEM_READY	
%M4	ALARM_INT	
%TM1.Q		DELAY_START

Rung1



Legend:

1 %TM1.Q

Variables used:

%M31	POWER_DELAY_ACTIVE	
%TM1.Q		DELAY_START

Rung2



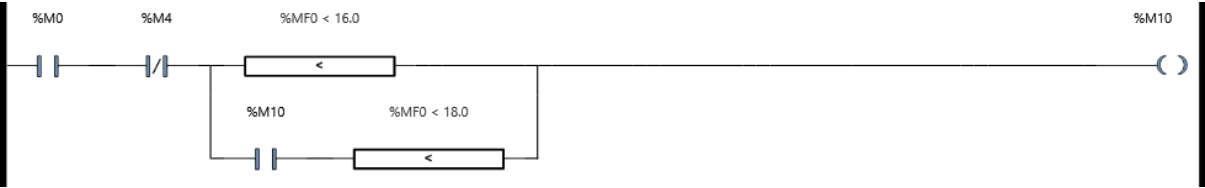
Variables used:

%M0	SYSTEM_READY	
%M4	ALARM_INT	
%M28	CIRC_PUMP_INT_CMD	

4 - POU_Modes

Master Task

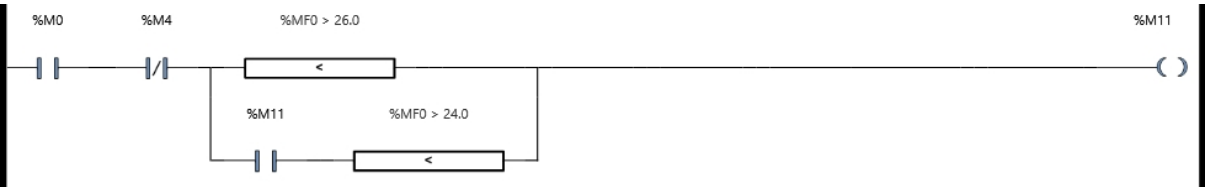
Rung0



Variables used:

%M0	SYSTEM_READY	
%M4	ALARM_INT	
%M10	HEATING_ACTIVE	
%MF0	ACTUAL_TEMP	Processed Outdoor Temp (REAL)

Rung1



Variables used:

%M0	SYSTEM_READY	
%M4	ALARM_INT	
%M11	COOLING_ACTIVE	
%MF0	ACTUAL_TEMP	Processed Outdoor Temp (REAL)

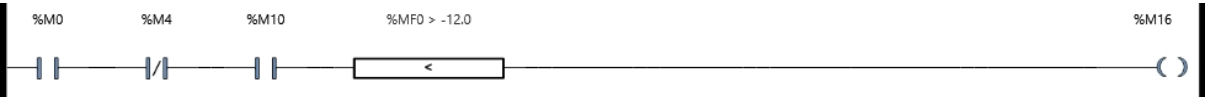
Rung2



Variables used:

%M0	SYSTEM_READY	
%M4	ALARM_INT	
%M10	HEATING_ACTIVE	
%M15	BOILER_MODE	
%MF0	ACTUAL_TEMP	Processed Outdoor Temp (REAL)

Rung3



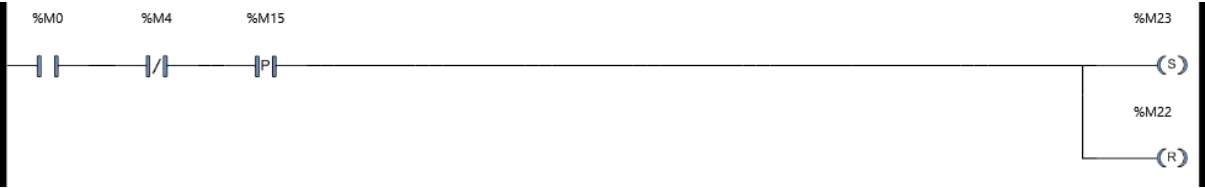
Variables used:

%M0	SYSTEM_READY	
%M4	ALARM_INT	
%M10	HEATING_ACTIVE	
%M16	CHILLER_MODE	
%MF0	ACTUAL_TEMP	Processed Outdoor Temp (REAL)

5 - POU_Valve_Logic

Master Task

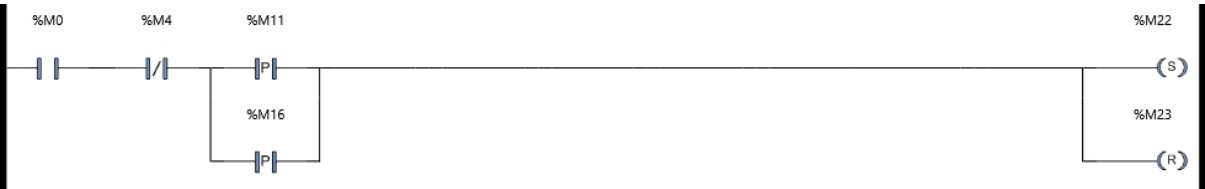
Rung0



Variables used:

%M0	SYSTEM_READY
%M4	ALARM_INT
%M15	BOILER_MODE
%M22	VALVE_TO_CHILLER
%M23	VALVE_TO_BOILER

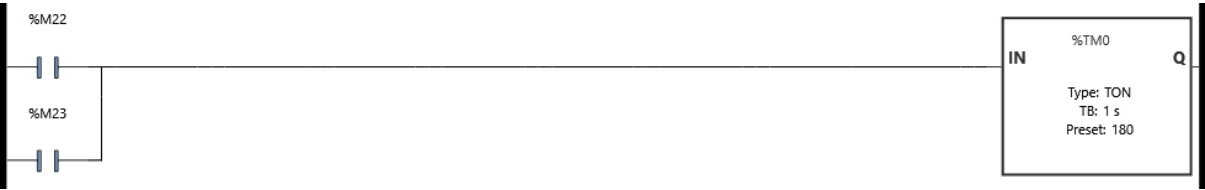
Rung1



Variables used:

%M0	SYSTEM_READY
%M4	ALARM_INT
%M11	COOLING_ACTIVE
%M16	CHILLER_MODE
%M22	VALVE_TO_CHILLER
%M23	VALVE_TO_BOILER

Rung2



Variables used:

%M22	VALVE_TO_CHILLER
%M23	VALVE_TO_BOILER
%TM0	

Rung3



Legend:

1 %TM0.Q

Variables used:

%M22 VALVE_TO_CHILLER

%M23 VALVE_TO_BOILER

%TM0.Q

Rung4



Variables used:

%M0 SYSTEM_READY

%M4 ALARM_INT

%M22 VALVE_TO_CHILLER

%M23 VALVE_TO_BOILER

Rung5



Variables used:

%M22 VALVE_TO_CHILLER

%M23 VALVE_TO_BOILER

%M24 VALVE_TIMER_ACTIVE

6 - POU_Outputs

Master Task

Rung0



Variables used:

%M28	CIRC_PUMP_INT_CMD	
%Q0.0	PUMP_START	Pump Control

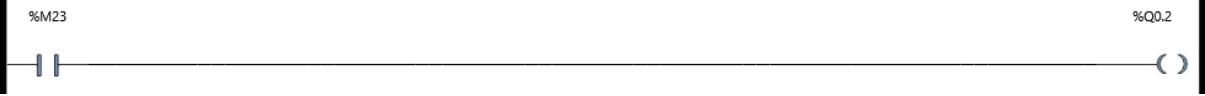
Rung1



Variables used:

%M0	SYSTEM_READY	
%M4	ALARM_INT	
%M10	HEATING_ACTIVE	
%M15	BOILER_MODE	
%M18	VALVE_AT_BOILER	
%Q0.1	BOILER_START	Daikin Remote ON/OFF(76-77)

Rung2



Variables used:

%M23	VALVE_TO_BOILER	
%Q0.2	VALVE_BOILER	Cool/Heat Selection(72-73)

Rung3



Variables used:

%M22	VALVE_TO_CHILLER	
%Q0.3	VALVE_CHILLER	Boiler Start

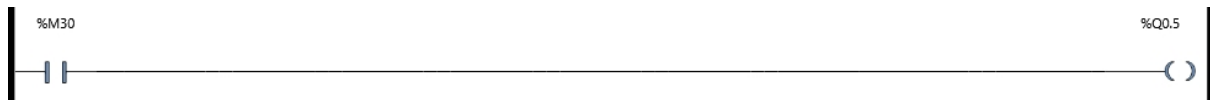
Rung4



Variables used:

%M0	SYSTEM_READY	
%Q0.4	LAMP_START	Move to Chiller Position

Rung5



Variables used:

%M30	DAIKIN_RUN_FB_INT	
%Q0.5	DAIKIN_RUN	Move to Boiler Position

Rung6



Variables used:

%M4	ALARM_INT	
%Q0.6	LAMP_ALARM	System Alarm Indicator

Rung7



Variables used:

%M0	SYSTEM_READY	
%M4	ALARM_INT	
%M5	DAIKIN_MODE_FB	
%Q0.7	LAMP_RUN	System Operation Indicator

Rung8



Variables used:

%M0	SYSTEM_READY	
%M4	ALARM_INT	
%M11	COOLING_ACTIVE	
%M16	CHILLER_MODE	
%M17	VALVE_AT_CHILLER	
%Q0.8	START_CHILLER	Heating Mode Active Indicator

Rung9



Variables used:

%M10	HEATING_ACTIVE	
%Q0.9	COOLING_HEATING	Cooling Mode Active Indicator

SYMBOLS

Used	Address	Symbol	Comment
X	%I0.0	MAIN_START	System Start Button
X	%I0.1	MAIN_STOP	Emergency Stop Button
X	%I0.2	CHILLER_PUMP_FB	pump feedback
X	%I0.3	DAIKIN_ALARM	
X	%I0.4	DAIKIN_RUN_FB	Fault Signal (80-81)
X	%I0.5	DAIKIN_MODE	Operation Feedback(82-83)
X	%I0.6	CIRC_PUMP_FB	Water Flow Signal(70-71)
X	%I0.7	DI_VALVE_FB	
X	%M0	SYSTEM_READY	
X	%M4	ALARM_INT	
X	%M5	DAIKIN_MODE_FB	
X	%M10	HEATING_ACTIVE	
X	%M11	COOLING_ACTIVE	
X	%M15	BOILER_MODE	
X	%M16	CHILLER_MODE	
X	%M17	VALVE_AT_CHILLER	
X	%M18	VALVE_AT_BOILER	
X	%M21	CHILLER_PUMP_FB_INT	
X	%M22	VALVE_TO_CHILLER	
X	%M23	VALVE_TO_BOILER	
X	%M24	VALVE_TIMER_ACTIVE	
X	%M27	PUMP_FB_INT	
X	%M28	CIRC_PUMP_INT_CMD	
X	%M30	DAIKIN_RUN_FB_INT	
X	%M31	POWER_DELAY_ACTIVE	
X	%MF0	ACTUAL_TEMP	Processed Outdoor Temp (REAL)
X	%MW2	TEMP_SENSOR_RAW	
X	%Q0.0	PUMP_START	Pump Control
X	%Q0.1	BOILER_START	Daikin Remote ON/OFF(76-77)
X	%Q0.2	VALVE_BOILER	Cool/Heat Selection(72-73)
X	%Q0.3	VALVE_CHILLER	Boiler Start

Used	Address	Symbol	Comment
X	%Q0.4	LAMP_START	Move to Chiller Position
X	%Q0.5	DAIKIN_RUN	Move to Boiler Position
X	%Q0.6	LAMP_ALARM	System Alarm Indicator
X	%Q0.7	LAMP_RUN	System Operation Indicator
X	%Q0.8	START_CHILLER	Heating Mode Active Indicator
X	%Q0.9	COOLING_HEATING	Cooling Mode Active Indicator
X	%S1	SB_WARMSTART	Indicates there was a warm start with data backup

CROSS-REFERENCE TABLE

Address	Object	Rung	Code
%I0.0....	3 - Main_Program	Rung0	-- P --
%I0.1....	3 - Main_Program	Rung0	-- --
%I0.2....	1 - POU_Inputs	Rung3	-- --
%I0.3....	1 - POU_Inputs	Rung2	-- --
%I0.4....	1 - POU_Inputs	Rung4	-- --
%I0.5....	1 - POU_Inputs	Rung6	-- --
%I0.6....	1 - POU_Inputs	Rung7	-- --
%I0.7....	1 - POU_Inputs	Rung8	-- --
		Rung9	-- / --
%IW0.100.	1 - POU_Inputs	Rung5	--[...]-- %MW2 := %IW0.100
%M0.....	3 - Main_Program	Rung0	--()--
			-- --
		Rung2	-- --
	4 - POU_Modes	Rung0	-- --
		Rung1	-- --
		Rung2	-- --
		Rung3	-- --
	5 - POU_Valve_Logic	Rung0	-- --
		Rung1	-- --
		Rung4	-- / --
	6 - POU_Outputs	Rung1	-- --
		Rung4	-- --
		Rung7	-- --
		Rung8	-- --
%M4.....	1 - POU_Inputs	Rung2	--()--
	3 - Main_Program	Rung0	-- / --
		Rung2	-- / --
	4 - POU_Modes	Rung0	-- / --
		Rung1	-- / --
		Rung2	-- / --
		Rung3	-- / --

Address	Object	Rung	Code		
	5 - POU_Valve_Logic	Rung0	-- / --		
		Rung1	-- / --		
		Rung4	-- --		
	6 - POU_Outputs	Rung1	-- / --		
		Rung6	-- --		
		Rung7	-- / --		
		Rung8	-- / --		
	%M5.....	1 - POU_Inputs	Rung6	--()--	
	6 - POU_Outputs	Rung7	-- --		
	%M10.....	4 - POU_Modes	Rung0	--()--	
			-- --		
		Rung2	-- --		
		Rung3	-- --		
	6 - POU_Outputs	Rung1	-- --		
		Rung9	-- --		
	%M11.....	4 - POU_Modes	Rung1	--()--	
				-- --	
			5 - POU_Valve_Logic	Rung1	-- P --
			6 - POU_Outputs	Rung8	-- --
		%M15.....	4 - POU_Modes	Rung2	--()--
5 - POU_Valve_Logic		Rung0	-- P --		
6 - POU_Outputs		Rung1	-- --		
%M16.....		4 - POU_Modes	Rung3	--()--	
		5 - POU_Valve_Logic	Rung1	-- P --	
			6 - POU_Outputs	Rung8	-- --
		%M17.....	1 - POU_Inputs	Rung8	--()--
	6 - POU_Outputs	Rung8	-- --		
	%M18.....	1 - POU_Inputs	Rung9	--()--	
	6 - POU_Outputs	Rung1	-- --		
	%M21.....	1 - POU_Inputs	Rung3	--()--	
	%M22.....	5 - POU_Valve_Logic	Rung0	--(R)--	
		Rung1	--(S)--		
		Rung2	-- --		
	Rung3	--(R)--			

Address	Object	Rung	Code
%M23.....	6 - POU_Outputs 5 - POU_Valve_Logic	Rung4	-- (R) --
		Rung5	-- --
		Rung3	-- --
		Rung0	-- (S) --
		Rung1	-- (R) --
		Rung2	-- --
		Rung3	-- (R) --
		Rung4	-- (R) --
		Rung5	-- --
		Rung2	-- --
%M24.....	5 - POU_Valve_Logic	Rung5	-- () --
%M27.....	1 - POU_Inputs	Rung7	-- () --
%M28.....	3 - Main_Program	Rung2	-- () --
%M30.....	6 - POU_Outputs	Rung0	-- --
	1 - POU_Inputs	Rung4	-- () --
	6 - POU_Outputs	Rung5	-- --
	%M31.....	1 - POU_Inputs	Rung0
		Rung1	-- --
3 - Main_Program		Rung1	-- (R) --
%MF0.....	2 - POU_Scaling	Rung0	--[...]-- %MF0 := ((INT_TO_REAL (%MW2) - 4000.0) / 16000.0) * 80.0 - 20.0
	4 - POU_Modes	Rung0	--[<]-- %MF0 < 16.0
			--[<]-- %MF0 < 18.0
		Rung1	--[<]-- %MF0 > 26.0
			--[<]-- %MF0 > 24.0
		Rung2	--[<]-- %MF0 < -15.0
	Rung3	--[<]-- %MF0 > -12.0	
%MW2.....	1 - POU_Inputs	Rung5	--[...]-- %MW2 := %IW0.100
	2 - POU_Scaling	Rung0	--[...]-- %MF0 := ((INT_TO_REAL (%MW2) - 4000.0) / 16000.0) * 80.0 - 20.0
%Q0.0....	6 - POU_Outputs	Rung0	-- () --
%Q0.1....	6 - POU_Outputs	Rung1	-- () --
%Q0.2....	6 - POU_Outputs	Rung2	-- () --
%Q0.3....	6 - POU_Outputs	Rung3	-- () --

Address	Object	Rung	Code
%Q0.4....	6 - POU_Outputs	Rung4	-- () --
%Q0.5....	6 - POU_Outputs	Rung5	-- () --
%Q0.6....	6 - POU_Outputs	Rung6	-- () --
%Q0.7....	6 - POU_Outputs	Rung7	-- () --
%Q0.8....	6 - POU_Outputs	Rung8	-- () --
%Q0.9....	6 - POU_Outputs	Rung9	-- () --
%S1.....	1 - POU_Inputs	Rung0	-- --
%TM0.....	5 - POU_Valve_Logic	Rung2	%TM0
%TM0.Q...	5 - POU_Valve_Logic	Rung3	-- --
%TM1.....	1 - POU_Inputs	Rung1	%TM1
%TM1.Q...	3 - Main_Program	Rung0	-- --
		Rung1	-- --